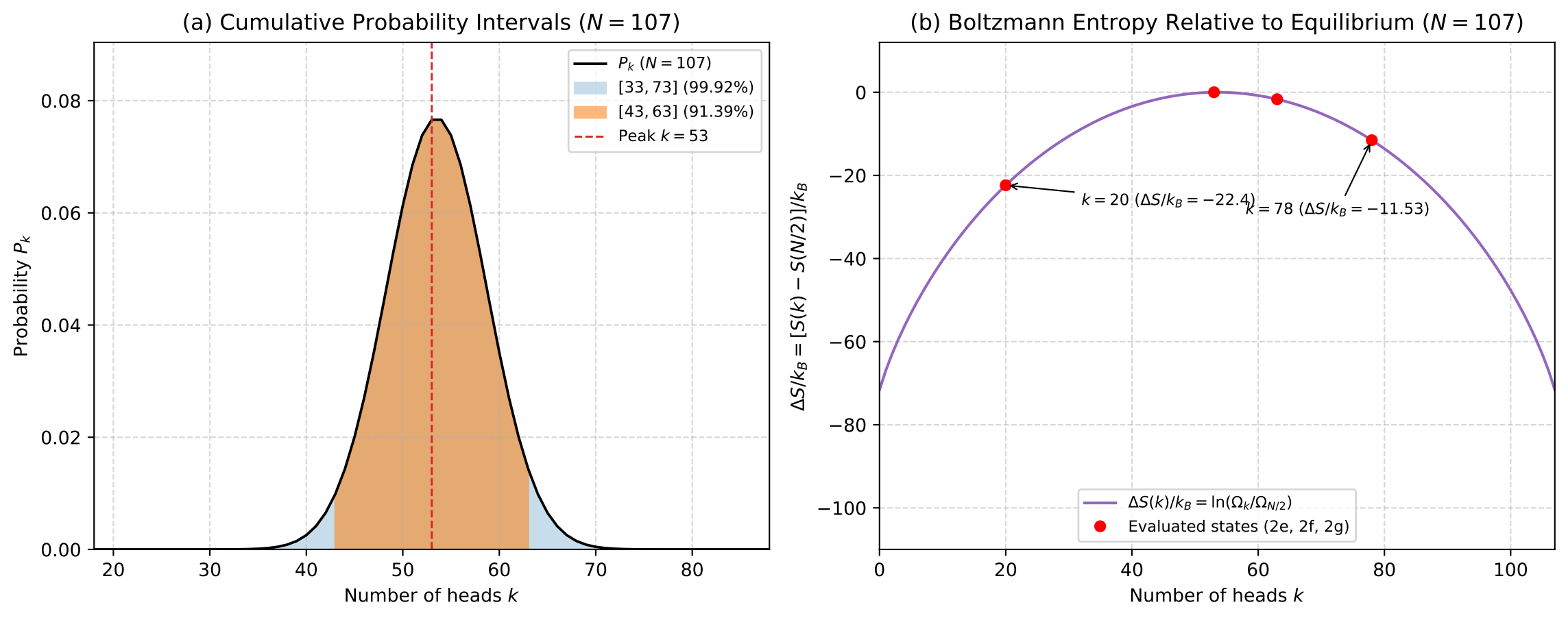
(a) Cumulative Probability Intervals ($N = 107$)
P_k ($N = 107$)
[33, 73] (99.92%)
[43, 63] (91.39%)
Peak $k = 53$
Probability P_k
Number of heads k
(b) Boltzmann Entropy Relative to Equilibrium ($N = 107$)
$k = 20$ ($\Delta S/k_B = -22.4$)
$k = 78$ ($\Delta S/k_B = -11.53$)
$\Delta S(k)/k_B = \ln(\Omega_k/\Omega_{N/2})$
Evaluated states (2e, 2f, 2g)
$\Delta S/k_B = [S(k) - S(N/2)]/k_B$
Number of heads k